DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

UNIVERSITY COLLEGE OF ENGINEERING (A)

Osmania University, Hyderabad-07

1005237290008

CLASS TEST-I

SUBJECT:SIGNAL AND SYSTEMs

Class: B.E. (AI&ML)

Semester: IV

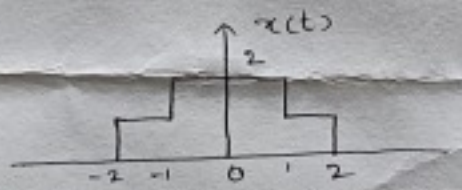
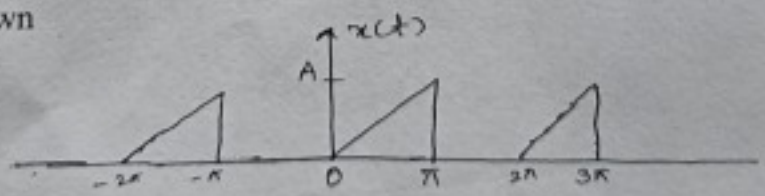
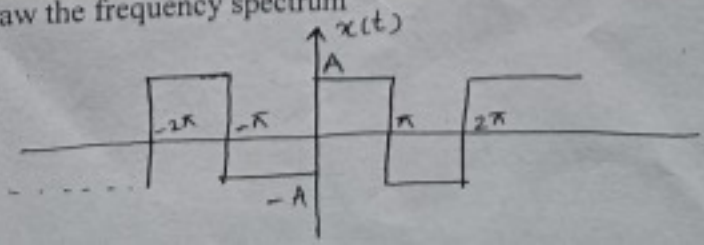
Academic Year: 2024-25

Max Marks: 30

Duration:1 hr 30 min

Date:29-03-2025

Answer All Questions

	PART A (5 x 2 = 10 M)	Marks	BT	CO
1. a)	State the Dirichlets conditions for the Fourier series to exist	2	2	2
b)	State two properties of the Fourier series	2	2	2
c)	Determine whether the signal x(t)=cos(2pi *t) cos(2πt)	2	2	1
d)	Distinguish between deterministic and random signals	2	1	1
e)	Define anti-causal signal	2	2	1
	Answer any two PART B (2 x 10 = 20 M)			
2.	(i) Determine the power,Energy of Asin(wt+θ)	5	2	1
	(ii) Find the energy of the signal shown	5	3	1
				
3. a)	For the given x[n]={1,4,3,-1,2} plot the following signals (i) x(-n-1) (ii) x(-n/2) (iii) x(-n/2)+2	5	3	1
b)	Obtain the trigonometric fourier series for the wave form shown 	5	3	2
4.a	Determine whether the system y(t) = 10x(t)+5 is static, linear,time invariant,casual and stable or not	4	4	1
4.b)	Obtain the exponential Fourier series for the wave form shown and also draw the frequency spectrum 	6	3	2